

## Exercise 1 - SQL Fundamentals

1 Retrieve all Columns from the employees table

```
SELECT *  
FROM employees;
```

| id | first_name | last_name | department | Salary | hire_date  | City         |
|----|------------|-----------|------------|--------|------------|--------------|
| 1  | Alice      | Green     | IT         | 70000  | 2020-01-10 | Johannesburg |
| 2  | Brian      | Lee       | HR         | 45000  | 2019-03-22 | Cape Town    |
| 3  | Cathy      | Zulu      | Finance    | 65000  | 2018-07-18 | Durban       |
| 4  | David      | Mokoena   | Marketing  | 50000  | 2021-11-05 | Pretoria     |
| 5  | Eva        | Naidoo    | IT         | 72000  | 2017-09-30 | Johannesburg |

2. Find all unique departments.

```
SELECT DISTINCT department  
FROM employees;
```

department  
IT  
HR  
Finance  
Marketing

3 Retrieve first and last names by Salary descending

```
SELECT first_name, last_name  
FROM employees  
ORDER BY Salary DESC;
```

| first_name | last_name |
|------------|-----------|
| Eva        | Naidoo    |
| Alice      | Green     |
| Cathy      | Zulu      |
| David      | Mokoena   |
| Brian      | lee       |

4. Retrieve the top 3 highest-paid employee

```
SELECT id, first_name, last_name, salary
FROM employees
ORDER BY salary DESC
LIMIT 3;
```

| id | first_name | last_name | Salary |
|----|------------|-----------|--------|
| 5  | Eva        | Naidoo    | 72000  |
| 1  | Alice      | Green     | 70000  |
| 3  | Cathy      | Zulu      | 65000  |

5. Find employees in the IT department

```
SELECT id, first_name, last_name, department
FROM employees
WHERE department = "IT";
```

| id | first_name | last_name | department |
|----|------------|-----------|------------|
| 1  | Alice      | Green     | IT         |
| 5  | Eva        | Naidoo    | IT         |

6. Find employees in finance with Salary > 60000

```
SELECT id, first-name, last-name, department, Salary  
FROM employees  
WHERE department = 'Finance' AND Salary > 60000;
```

| id | first-name | last-name | department | Salary |
|----|------------|-----------|------------|--------|
| 3  | Cathy      | Zulu      | Finance    | 65000  |

7 Find employees in HR or Marketing

```
SELECT id, first-name, department  
FROM employees  
WHERE department = 'HR' OR department = 'marketing';
```

| id | first-name | last-name | department |
|----|------------|-----------|------------|
| 2  | Brian      | lee       | HR         |
| 4  | David      | Mokoena   | Marketing  |

8 Find employees not in IT

```
SELECT id, first-name, last-name, department  
FROM employees  
WHERE department != 'IT';
```

| id | first-name | last-name | department |
|----|------------|-----------|------------|
| 2  | Brian      | lee       | HR         |
| 3  | Cathy      | Zulu      | Finance    |
| 4  | David      | Mokoena   | Marketing  |

09 Find employees in IT, HR or FINANCE using IN

```
SELECT id, first_name, last_name, department
FROM employees
WHERE department IN ('IT', 'HR', 'finance');
```

| id | first_name | last_name | department |
|----|------------|-----------|------------|
| 1  | Alice      | Green     | IT         |
| 2  | Brian      | lee       | HR         |
| 3  | Cathy      | Zulu      | Finance    |
| 5  | Eva        | Naidoo    | IT         |

10 Find employees in IT with Salary > 65000 and city Johannesburg

```
SELECT id, first_name, last_name, department, Salary, city
FROM employees
WHERE department = 'IT' AND Salary > 65000 AND city = 'Johannesburg'
```

| id | first_name          | last_name | department | Salary | city         |
|----|---------------------|-----------|------------|--------|--------------|
| 1  | Alice               | Green     | IT         | 70000  | Johannesburg |
| 5  | <del>Emma</del> Eva | Naidoo    | IT         | 72000  | Johannesburg |